

Magnetic switch Replacement

1. The Problem

The SafeStep wristband uses a mechanical tact switch (E-Switch TL3342F160QG) for the SET button. Short-press initiates BLE pairing; long-press enters configuration mode. Because the wristband is worn 24/7 by dementia patients — who frequently scratch, brush, or pick at unfamiliar objects on the body — the switch is fully vulnerable to accidental activation. A press during pairing mode disconnects the device from the dongle and silences the alarm, defeating the system's primary safety function.

A switch technology that cannot be triggered by mechanical contact is required. Two reed switches from Meder Electronic (MMS-1A-V2 and MMS-B-2).

2. Comparison

Property	Meder MMS-1A-V2	Meder MMS-B-2 ✓
Activation method	Magnetic field	Magnetic field
Accidental-press immunity	✓ Complete	✓ Complete
Quiescent current	0 (passive)	0 (passive)
Package size	1.30 × 0.90 mm	4.80 × 2.05 mm
Hand-assembly	Very difficult	Easy
Documented PCB pads	No	Yes
Switching life (cycles)	10 ⁶	10 ⁷
Unit cost	≈ €0.40	≈ €0.40
Schematic complexity	Same (passive)	Same (passive)

Why the MMS-B-2 is the best fit

- Eliminates the accidental-press vulnerability completely (only a magnet can activate it).
- Zero quiescent current — preserves the tight CR2032 battery.
- Standard SMD size — hand-solderable for prototypes, unlike the sub-mm MMS-1A-V2.
- Passive contact — the schematic is essentially unchanged from the existing design.

3. Selected Part — Meder MMS-B-2

Attribute	Value
Manufacturer	Meder Electronic (Standex group)
Part number	MMS-B-2 (Meder item No. 9210000022)
Package	4.80 × 2.05 × 1.39 mm SMD, tape & reel (500 pcs)
Pull-in field (operate)	1.0 – 4.0 mT
Drop-out field (release)	0.5 mT minimum
Contact rating	0.3 mW maximum

Attribute	Value
Switching current	100 μ A maximum
Operate time (incl. bounce)	0.1 ms typ, 1 ms max
Life expectancy	10^7 cycles at 1.5 V / 15 μ A
Shock / vibration	5000 g shock, 30 g vibration
Operating temperature	-20 to +100 °C
Distributors (Germany)	Mouser DE, Digi-Key DE

4. Activation Magnet

Activation Magnet

Property	Value
Material / grade	NdFeB, N42 or higher
Shape and size	Disc, 3 mm diameter $\times$ 1 mm thickness
Operating air gap	2 – 5 mm
Orientation	N–S axis parallel to the reed switch long axis

5. Accidental Trigger Mitigation

The reed switch eliminates accidental mechanical presses but can still be triggered by stray magnetic fields (wireless chargers, magnetic bag clasps, earbuds at close contact).

6. Conclusion

After evaluating the original mechanical tact switch (TL3342F160QG) against the two reed switch's (Meder MMS-1A-V2 and MMS-B-2), the **Meder MMS-B-2 reed switch is selected as the best option for the SafeStep.**

It is the only switch that satisfies all four high-priority project requirements simultaneously: complete immunity to accidental mechanical press by the dementia patient, zero quiescent current to preserve the CR2032 battery and minimal disruption to the existing KiCad schematic. And a small 3×1 mm N42 NdFeB magnet on the activation cap.

The MMS-B-2's standard SMD package (4.80×2.05 mm) makes it hand-solderable for prototyping, and its 1.0 – 4.0 mT pull-in range comfortably accommodates a standard small magnet at a 3 – 5 mm air gap.